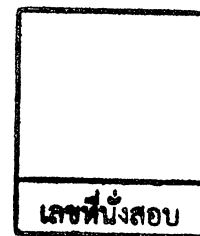


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 1

Year: 2017

Subject: 394173 Mathematics III

Section: 15-16

Date: 29 November 2017

Time: 13.00-16.00

Name _____ ID _____ Class _____

Instructions

1. The examination has 9 pages, 8 questions and a total score of 39 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are **NOT** allowed in the examination.
9. Electronic communication devices are **NOT** allowed in the examination room.

**Cheating in the exam is considered an extremely serious
offence which will result in expulsion from the University.**

No.	1.	2.	3.	4.	5.	6.	7.	8.	Total
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1. Assume that $k - 2$, $k - 6$, $2k + 3$ is the geometric sequence, where $k > 0$ and $-6, x, y, z, 8$ is the arithmetic sequence which $a_w = 64$. Determine $(w - x - y - z)^k$ (4 points)

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2. Find each limit, or state that the limit does not exist and explain your reasoning.

2.1 $a_n = \left(\frac{3n^4 + 5n^3 - 3n - 5}{15n^4 - 7n} \right) + (\sqrt{n^2 - 9} - \sqrt{n^2 - 1})$ (2 points)

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2.2 $a_n = (-1)^{n+3} \frac{3^n + 3 \cdot 9^n + 2 \cdot 27^n}{3^{3n} \cdot 54}$ (2 points)

3. Let $\sum_{n=1}^4 (bn^3 + 10) = 540$ (4 points)

Find the sum of first b terms of $3 \cdot 1 + 6 \cdot 5 + 9 \cdot 9 + 12 \cdot 13 + \dots$ (Use $\sum$)

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4. If the sum of first m terms of $5 + 12 + 19 + \dots = 1430$

and $\frac{3^{3m} + 3^{3m-1} + 3^{3m-2} + \dots + 3^{16}}{3^{17}} = \frac{3^a - 1}{b}$. Find $a + b$.

(4 points)

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5. Given $S = 1 + \frac{2}{5} + \frac{3}{25} + \frac{4}{125} + \dots$ and $16S - n = \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \dots$

Find the value of n .

(5 points)

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6. In a quadrilateral $ABCD$, $A(-5, 2)$, $B(-7, -2)$, $C(2, 5)$ and $D(4, -1)$
where E, F is the midpoint of AB, CD respectively

Find the vector has length as $|\vec{EF} - \vec{BC}|$ which has the same direction as $\vec{BC} - \vec{AD}$.
(4 points)

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7. Assume that $\vec{u}$, $\vec{v}$ and $\vec{w}$ are the vectors such that $\vec{u} + \vec{v} + \vec{w} = \vec{0}$ and $|\vec{u}| = 1$, $|\vec{v}| = 4$, $|\vec{w}| = 5$. Find $3|\vec{u} - \vec{v}| - 2|\vec{v} - \vec{w}|$? (4 points)

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- 8.1 Find the volume of the parallelepiped where (3 points)
 $\vec{u} = 3\hat{i} + \hat{j} + 2\hat{k}$, $\vec{v} = 4\hat{i} + 5\hat{j} + \hat{k}$ and $\vec{w} = \hat{i} + 2\hat{j} + 4\hat{k}$.

8.2 Suppose that $\vec{u} = 4\hat{i} - 3\hat{j} + 5\hat{k}$, $\vec{v} = -3\hat{i} - 5\hat{j} - 2\hat{k}$.

(7 points)

Find a) $\vec{w}$ satisfy condition $\vec{u} - 3\vec{v} + 2\vec{w} = \vec{0}$

b) $(\vec{u} \cdot \vec{w})(\vec{u} \times \vec{v})$

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